



VASIREDDY VENKATADRI INSTITUTE OF TECHNOLOGY

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Accredited by NAAC with 'A' Grade, ISO 9001:2008 Certified

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Department of CSE - Artificial Intelligence & Machine Learning

MAJOR PROJECT REGISTRATION FORM

Date: 31-07-2026

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	AIML - C5
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Multi-Agent AI Business Assistant for Intelligent Enterprise Workflow Automation

Abstract

Modern enterprise operations require seamless coordination across multiple functional domains, including Human Resources (HR), Finance, Software Engineering, Email Communication, Data Analytics, and Customer Support. However, traditional single-agent conversational AI systems struggle to execute complex cross-functional workflows due to limited contextual understanding, constrained context windows, and the absence of domain-specific reasoning and tool integration. Business processes such as employee onboarding, financial reconciliation, software provisioning, report generation, and customer communication often require collaboration across multiple departments, leading to delays and increased operational overhead. Existing automation approaches, including rule-based Robotic Process Automation (RPA), lack adaptability to dynamic enterprise scenarios, while current multi-agent frameworks such as AutoGen, CrewAI, and MetaGPT provide collaborative capabilities but offer limited support for enterprise-grade security, compliance, auditability, and coordinated workflow execution. To address these challenges, this paper proposes a Multi-Agent AI Business Assistant, an intelligent collaborative architecture designed specifically for enterprise workflow automation. The proposed system consists of six specialized AI agents: an HR Agent, Finance Agent, Coding Agent, Email Agent, Data Analysis Agent, and Customer Support Agent, coordinated by a centralized Coordinator Agent and an intelligent Intent Routing Engine. Each specialized agent handles domain-specific tasks using dedicated knowledge, reasoning strategies, and integrated enterprise tools, while the Coordinator Agent manages task allocation, synchronization, dependency resolution, and response aggregation. The architecture decomposes complex user requests into domain-specific subtasks, enables asynchronous inter-agent communication through an event-driven message bus, integrates enterprise tools such as payroll calculators, SQL query generators, code analysis utilities, email services, and analytical modules, and aggregates intermediate results into a unified response while maintaining structured execution logs for transparency and compliance. The proposed system focuses on secure digital enterprise workflow automation, intelligent task orchestration, and scalable cross-domain collaboration, aiming to improve task completion accuracy, reduce workflow execution time, enhance operational efficiency, and provide a reliable, auditable, and extensible AI-driven business assistant for modern enterprises. The proposed framework also emphasizes scalability, fault tolerance, and secure communication among agents, making it suitable for deployment in real-world enterprise environments. Through coordinated decision-making and seamless integration with enterprise applications, the system minimizes manual intervention while ensuring consistent, transparent, and reliable workflow execution.



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References (Category specific)

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- [5] "Agentic Artificial Intelligence as a New Frontier in Information Systems: Promise, Peril, and Research Opportunities," Information & Management, 2026. (Enterprise applications, opportunities, and challenges of Agentic AI in information systems.)
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